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THE SPECTRAL THEORY OF SEISMIC SOURCES

BY M. J. RANDALL

ABSTRACT

The far-field results of Brune's spectral theory are shown to be largely independent of his source model; this implies that the theory has even greater power than it seemed, but that its success in explaining the observed spectra does not in itself justify a dislocation model. Expressions are derived for seismic energy and characteristic stress which are independent of assumptions as to source model. For several models, the characteristic stress is found to be a good approximation to the stress-drop.

A theoretical relationship between earthquake magnitude M_L, stress-drop, and fault size is derived. This provides a means of estimating the stress-drop for earthquakes associated with aftershock sequences, using estimates of M_L and fault size, and is consistent with empirical relationships between magnitude and fault size, and between seismic energy and magnitude.

INTRODUCTION

Since the seminal work of Brune (1970), there has been a veritable explosion of papers on the seismic spectrum and its relationship to source size, seismic moment, and stress-drop. This bears convincing testimony to the power of his theory. Observations of spectra consistent with it have been reported for microearthquakes with very low stress-drops (Douglas and Ryall, 1972) and high stress-drops (Tucker and Brune, 1973), as well as for large earthquakes (Hanks and Wyss, 1972; Wyss and Hanks, 1972), thus covering a very wide range of the parameters of the theory.

Brune considers the physics of the situation when traction on a circular fault in a shear field is suddenly lost. The elastostatic problem was considered by Keilis-Borok (1959), who obtained relationships between stress-drop, seismic moment, and fault size. In principle, the seismic spectrum requires the dynamical solution of the problem, which so far is lacking. It is Brune's triumph that by physical argument he has obtained the maximum information on the nature of the spectrum without having a complete dynamical solution. The aim of this paper is in part to show, by considering other models (without implying that they are better models of the seismic source or even as good), that some principal results of Brune's theory are not strongly dependent on his model. This improves the power of his theory to provide reliable estimates of seismic energy and stress-drop, but implies, on the other hand, that the success of Brune's theory in describing the observations of seismic spectra does not in itself indicate that dislocation is the seismic source. Considering the weight of evidence for dislocation, this is a minor loss. The remainder of this paper proceeds in the direction first explored by Aki (1967) and deduces theoretical relationships between local magnitude M_L, stress-drop, and source size. To the existing techniques for estimating stress-drop, using field observations of fault offset and size on the one hand, and seismic moment and corner frequency on the other, is added a technique requiring observations of M_L and fault size. This has been usefully employed to estimate stress-drop in large earthquakes from the fault area as indicated by the spatial range of aftershocks (Gibowicz, 1973).

To facilitate comparisons between observation and theory, seismic sources are usually considered to be situated in an infinite homogeneous elastic region. Observations of wave

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displacement or its spectrum are "corrected" or referred back to a virtual focal sphere about the origin. "Correction" involves appropriate allowances for geometric and anelastic attenuation and for the effects of the crust and free surface, as well as for any instrumental effects.

We shall be concerned here only with the "wave term" of the dynamic displacement field. In general, the dynamic displacement field has two principal terms, one whose amplitude varies inversely with distance from the source and which is temporary; i.e., nonzero only in a limited range of time. The other term varies inversely as the square of distance and carries the permanent displacement field. It is analogous to the "induction field" in electromagnetic wave theory. The two terms are not unrelated, and some care is necessary in their consideration, especially if asymptotic approximations are made in the frequency domain. The wave term should be isolated first by taking the limit $r \to \infty$; any limits involving time or frequency should be taken subsequently. The normalization to be used here for the amplitude spectrum $\Omega(\omega)$ of wave displacement $u(t)$ is implicit in the defining equation

$$\Omega(\omega) = \int_{-\infty}^{\infty} u(t)\, e^{-i\omega t}\, dt.$$

Asymptotic Behavior of Seismic Spectra

Elastic multipole theory gives a relationship between the zero-frequency limit of the amplitude spectrum of wave displacement and seismic moment. For the P wave this takes the form

$$\Omega_p(O) = \frac{1}{4\pi r} \frac{M_o}{\mu} \frac{\beta^2}{\alpha^3} \mathscr{R}_p(\theta, \phi). \tag{1}$$

$\mathscr{R}_p(\theta, \phi)$ is a factor allowing for the radiation pattern. The factor for the classical dislocation source, when referred to spherical polar coordinates having the conventional relationship to a cartesian frame with the fault plane $z = 0$ and slip in the y direction, is

$$\mathscr{R}_p = \sin 2\theta \sin \phi.$$

This relationship was derived for the multipolar point source by Keilis-Borok *et al.* (1960); its generalization has been discussed by Randall (1971a, b).

For the S wave the analogous relationship is

$$\Omega_S(O) = \frac{1}{4\pi r} \frac{M_o}{\mu} \frac{1}{\beta} \mathscr{R}_S(\theta, \phi). \tag{2}$$

The (vector) factor that allows for the radiation pattern takes the form

$$\mathscr{R}_S = (0, \cos 2\theta \sin \phi, \cos \theta \cos \phi)$$

in polar coordinates for the classical dislocation source. Both expressions are quite general, the factor M_o/μ being simply generalized to the shear invariant L (Randall, 1971b) for other hypothetical sources.

At the other extreme of the spectrum, the high-frequency asymptotic behavior of $\Omega(\omega)$ is related to the singularities of the time-function $u(t)$. Lighthill (1958) shows that for asymptotic behavior of the form $\Omega \sim \omega^{-\gamma}$, γ being some constant, the "worst" singularity in $u(t)$ is of the form

$$u(t) \sim |t - t_o|^{\gamma - 1}.$$

In order that the total energy should remain finite in the spectral estimate afforded by the application of Parseval's theorem, γ must be greater than 1.5 (Hanks and Wyss, 1972).

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Furthermore, if $\gamma < 2$ the singularity in the time function would involve an infinite discontinuity in velocity. On the other hand, $\gamma = 2$ corresponds to a "worst" singularity of type $|t-t_o|$, which has a finite jump in velocity. This is exactly what is to be expected from a relaxation source, for a wave front carrying a finite jump in velocity is a consequence of the sudden appearance of a stress discontinuity. Examples are discussed by Love (1904), Randall (1966), and Brune (1970). Noninteger values of $\gamma > 2$ would involve time functions with branch-point singularities and are not to be expected in physical situations. Thus, there are sound theoretical reasons for expecting ω^{-2} behavior at high frequencies in seismic spectra.

The amplitude spectrum can thus be expressed in the form

$$\Omega(\omega) = \Omega(O)\, F(x)$$

where $F(x)$ is a function of dimensionless frequency x, such that $F(O) = 1$ and $F \sim x^{-2}$ at high frequencies. As it turns out, F may oscillate at high frequency with an upper bound $\sim x^{-2}$. As a point of convention, it is proposed that $x(= \omega/2\pi\nu$, say) be chosen so that the upper bound of $x^2F(x) \to 1$ as $x \to \infty$. This affords a definition of "corner frequency" ν which differs a little from the usual definition when $F(x)$ oscillates at high frequency, but as it is operationally convenient, especially when it comes to the estimation of corner frequency for an observed spectrum contaminated by noise, its use may be recommended.

The precise functional form of $F(x)$, and the relationship of corner frequency to source size, will depend upon the model assumed for the seismic source mechanism. The source theory of Randall (1964) predicts, for relaxation models, wave potentials directed steadily toward the final equilibrium configuration, with as a consequence, displacement pulses that do not change sign. Thus,

$$|\Omega(\omega)| = \left| \int_{-\infty}^{\infty} u(t)\, e^{-i\omega t}\, dt \right| \leqq \int_{-\infty}^{\infty} |u(t)|\, dt = |\Omega(O)|.$$

So, while $|\Omega(\omega)|$ might oscillate with increasing frequency, it does so with an upper bound which is less than its value at zero-frequency, and which decreases as ω^{-2} asymptotically. The corner frequency defined here corresponds to the intersection of the low-frequency asymptote $F = 1$ and the high-frequency asymptote of the upper limit $F = 1/x^2$.

Spectral Shape, Energy and Stress

One example of a model of earthquake mechanism for which a sufficiently complete solution is known to give $\Omega(\omega)$ explicitly is that of Randall (1966), which discusses the radiation from an initial state with a pre-stressed spherical inclusion in an unstressed medium. This model yields parabolic displacement pulses for both P and S waves and the spectral shape function is

$$F(x) = 3^{-1/2}\, x^{-3}\, [\sin(3^{1/2}x) - 3^{1/2}x \cos(3^{1/2}x)] \tag{3}$$

where $x = \omega/2\pi\nu = 3^{-1/2}\omega R/c$, R is the source radius and c is the appropriate seismic velocity. Another model yielding this spectral shape is that of Archambeau (1968), which involves the sudden vanishing of stress inside a sphere; the correct wave spectrum for this model has been calculated by Randall (1973).

An analogous solution for a dislocation model is not yet available. The simplest dislocation model that would serve the purpose is probably that which involves the sudden loss of traction across a fault surface in a stressed region, or alternatively, one that involves the sudden application of opposing tractions to the two faces of a fault in an unstressed medium. The corresponding static problem for a disc-shaped fault has been solved by

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Keilis-Borok (1959), yielding expressions for the displacement dislocation and seismic moment, in terms of the applied stress (or stress-drop). Without the dynamical solution, however, it is not possible to deduce the spectral shape function $F(x)$ explicitly.

There is, however, an alternative approach to the problem that yields useful results. First, one can assume a plausible spectral shape function (bearing in mind that $F(O) = 1$, $x^2F(x) \to 1$ at high frequency). Brune (1970), for example, assumes $F(x) = (1+x^2)^{-1}$. One might go on to deduce the value of the constant of proportionality, k, in the relationship between corner frequency and fault radius

$$2\pi\ \nu = kc/R \tag{4}$$

by the requirement that energy be conserved. Alternatively, k might be evaluated in some other way and the available potential energy be compared with the radiated energy that corresponds to the assumed $F(\mu)$, thus permitting some estimate of the way in which this first approximation differs from the true spectral shape function. Brune (1970) argued that conservation of energy density in the wave front (which is associated with the ω^{-2} high-frequency behavior) required a value of 2.21 for k [later corrected to 2.34, Brune (1971)]. This value gives too little radiated energy with his assumed $F(x) = (1+x^2)^{-1}$; thus leading Brune to suggest that the true spectral amplitudes are a little higher in the neighborhood of the corner frequency. Figure 1 shows the spectral shape implicit in the

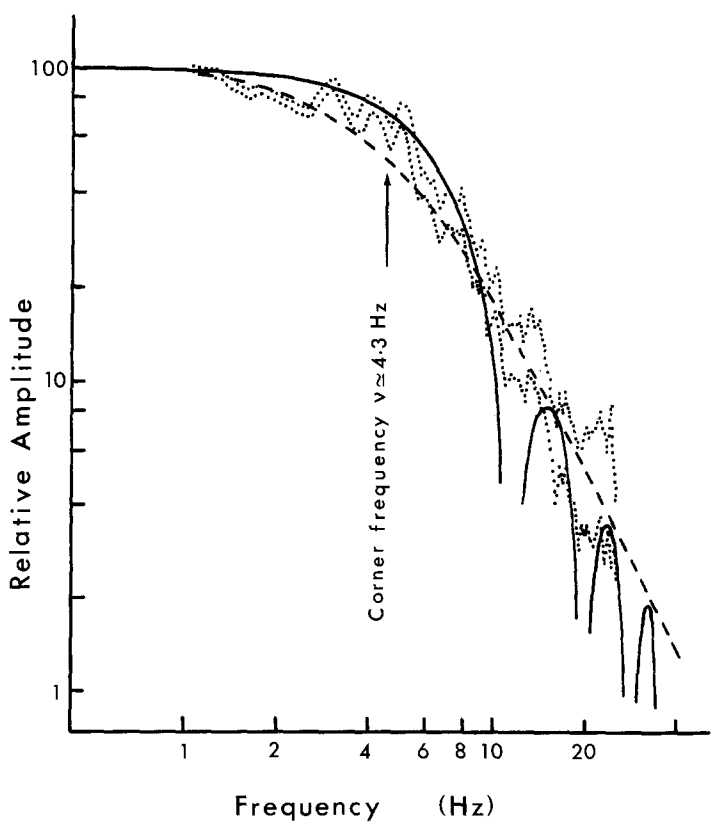

FIG. 1. Spectral shape functions of Randall (1966) (solid curve) and Brune (1970) (dashed). The dotted curves outline the likely range of average spectral amplitude estimated by Douglas and Ryall (1972) for a number of microearthquakes in Nevada. The corner frequencies of the theoretical curves have been adjusted to that estimated by Douglas and Ryall.

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model of Randall (1966) and that assumed by Brune, and compares them with a particularly well-defined average spectrum observed by Douglas and Ryall (1972) for a number of microearthquakes near Fairview Park, Nevada. Randall's spectral amplitude is indeed higher near the corner frequency and seems to fit the observations a little better.

Expressing the spectral amplitudes for either the P wave or the S wave in the general form

$$\Omega(\omega) = \frac{M_o}{4\pi r\, \rho\, c^2}\, \mathscr{R}(\theta, \phi)\, F(x); \quad x = \omega/2\pi\, v \tag{5}$$

and further assuming that v is a function only of the source radius R, we may evaluate the energy radiated in either wave by applying Parseval's theorem. This yields

$$E_R\,(P \text{ or } S) = \tfrac{1}{2}\langle \mathscr{R}^2 \rangle\, \rho^{-1}\, c^{-5}\, M_o^{\,2}\, v^3 \textstyle\int_0^\infty x^2\, F^2(x)\, dx. \tag{6}$$

The ratio of energies in the P and S waves is, therefore,

$$\frac{E_R(S)}{E_R(P)} = \frac{\langle \mathscr{R}_s^{\,2} \rangle}{\langle \mathscr{R}_p^{\,2} \rangle} \frac{\alpha^2}{\beta^2} = \frac{3}{2} \frac{\alpha^2}{\beta^2}$$

which equals 9/2 for a Poisson solid. Thus, the total radiated energy is

$$E_R = \frac{11}{9} \frac{4\pi}{5} \frac{M_o^{\,2}}{\rho} \frac{v_s^{\,3}}{\beta^5} \int_0^\infty x^2\, F^2(x)\, dx, \tag{7}$$

this relation being independent of any choice of model. For a dislocation model in which the tangential traction on a circular fault drops suddenly from σ_1 to $\sigma_2 = \sigma_1 - \sigma$, the energy available for seismic radiation is (Randall, 1972)

$$W = \frac{1}{2}\, \sigma\, M_o/\mu. \tag{8}$$

Keilis Borok's (1959) result for such a model is

$$M_o = \frac{16}{7}\, \sigma\, R^3. \tag{9}$$

Thus, with

$$2\pi\, v_s = k\, \beta/R$$

$$W = \frac{7\pi^3}{4\rho} \frac{M_o^{\,2}}{\beta^5} \frac{v_s^{\,3}}{k^3}. \tag{10}$$

Conservation of energy, therefore, requires that

$$k^3 \doteqdot 17.3/ \textstyle\int_0^\infty x^2\, F^2(x)\, dx. \tag{11}$$

The likely range of values for the integral

$$I_F = \textstyle\int_0^\infty x^2\, F^2(x)\, dx$$

is exemplified by consideration of four particular spectral shape functions

1. A function equated to the low- and high-frequency asymptotes intersecting at the corner frequency

$$F(x) = 1,\ 0 \leqq x \leqq 1$$
$$= x^{-2}, \quad x > 1.$$

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For this function

$$I_F = \frac{4}{3} (\doteqdot 1.33)$$

and

$$k \doteqdot 2.35.$$

2. The function considered by Brune (1970)

$$F(x) = (1+x^2)^{-1}$$

for which

$$I_F = \frac{\pi}{4} (\doteqdot 0.79)$$

and

$$k \doteqdot 2.80.$$

3. The spectral shape for either of the spherical source models of Randall (1966) and Archambeau (1968)

$$F(x) = 3^{1/2}\, x^{-3}\, [\sin (3^{1/2}x) - 3^{1/2}x \cos (3^{1/2}x)]$$

for which

$$I_F = \frac{1}{2} \pi\, 3^{-1/2} (\doteqdot 0.91)$$

and

$$k \doteqdot 2.67.$$

4. A function defined asymptotically like function 1 but assuming a ω^{-3} high-frequency behavior

$$F(x) = 1,\ 0 \leqq x \leqq 1$$
$$= x^{-3}, \quad x > 1$$

for which

$$I_F = \frac{2}{3} (\doteqdot 0.67)$$

and

$$k \doteqdot 2.96.$$

Thus, with the possible exception of cases involving very low fractional stress-drop [$\varepsilon \ll 1$ in the notation of Brune (1970), or very elongated faults (Savage, 1972)], I_F is about unity and, to an adequate degree of precision, the total seismic energy can be estimated [using equation (7)], given estimates of the seismic moment M_o and the corner frequency ν_s (defined as the intersection of the low-frequency and the high-frequency asymptotes) by the formula

$$E_R \doteqdot 3.0\, \frac{M_o^{\,2}\, \nu_s^{\,3}}{\rho\, \beta^5}, \tag{12}$$

without appealing to a particular source model.

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By analogy with the general model of stress-drop discussed above, which satisfies equation (8), we may define a "characteristic stress"

$$\sigma = 2\mu\, E_R/M_o$$
$$\doteqdot 6.0\, \frac{M_o\, v_s^{\,3}}{\beta^3}. \tag{13}$$

This stress corresponds to twice the "apparent stress" (Wyss and Molnar, 1972) when E_R is estimated from earthquake magnitude using the Gutenberg and Richter relationship. It is, of course, exactly equal to the stress-drop for those dislocation models in which the stress drops instantaneously to its final value; the situation is more complicated when the final stress exceeds the stress operative during relative motion of the faces of the fault ("partial stress-drop" in the terminology of Brune (1970)). For the model involving loss of traction on a sphere (Archambeau, 1968), this characteristic stress also exactly equals the stress-drop (Aki and Tsai, 1972). Similarly, when account is taken of the residual stress in the inclusion, Randall's (1966) model has stress-drop

$$\Delta\sigma = \frac{1}{15}\,\frac{9\lambda+14\mu}{\lambda+2\mu}\,k'$$

where k' is the initial shear strain in the inclusion; the radiated energy is

$$E_R = E_p + E_s$$
$$= \frac{2}{15}\,\mu\, k^2\, V\,\frac{\beta^2}{\alpha^2} + \frac{1}{5}\,\mu\, k^2\, V$$
$$= \frac{1}{15}\,\frac{3\lambda+8\mu}{\lambda+2\mu}\,k'\, M_o$$

and the characteristic stress is

$$\sigma = f\left(\frac{\lambda}{\mu}\right)\Delta\sigma$$

where

$$f(\lambda/\mu) = \frac{2(3\lambda+8\mu)}{9\lambda+14\mu} = \frac{22}{23}$$

for a Poisson solid. Thus the characteristic stress defined here is again almost equal to the stress-drop.

If the complete dynamical solution for the dislocation source should involve a corner frequency that depends also on the angle at which the ray leaves the source, v_s in equations (12) and (13) must be regarded as an average corner frequency.

Earthquake Magnitude M_L

Given the theoretical spectral amplitude, the local magnitude M_L that is to be expected may easily be calculated. If, as is sufficiently accurate for the present purpose, we assume a spectral function for the S wave

$$\Omega(\omega) = \Omega(O)\,[1+(\omega/2\pi\, \nu)^2]^{-1}$$

where ν, the (average) corner frequency, is related to source size by the formula

$$2\pi\, \nu = 2.34\beta/R$$

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and $\Omega(O)$ is defined in terms of seismic moment

$$\Omega(O) = \frac{1}{4\pi r}\frac{M_o}{\mu\beta}$$

which, in turn, may be expressed in terms of stress-drop and fault size

$$M_o = \frac{16}{7}\,\sigma\,R^3,$$

we may evaluate the displacement spectrum in terms of stress-drop and fault size. By multiplying this spectrum by the spectral response of a standard Wood-Anderson torsion seismograph

$$|S(\omega)| = \left|\frac{\omega^2}{\omega^2+2ih\,\omega\,\omega_0-\omega_0{}^2}\right|,$$

$$\omega_0 \doteqdot 7.9\ \text{sec}^{-1},$$

$$h \doteqdot 0.8,$$

and allowing a factor of 2 for the effect of the free surface, we may estimate the maximum amplitude that would be registered on such a seismograph at a distance of 100 km and hence obtain the local magnitude

$$M_L = \log A/A_o,$$

$$A_o = 10^{-4}\ \text{cm}.$$

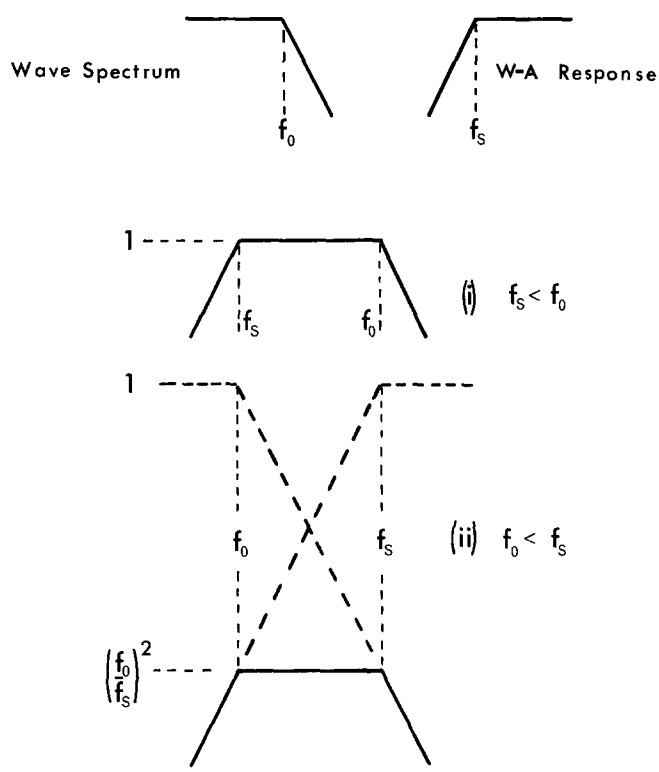

FIG. 2. Schematic spectra of the displacement wave and the response Wood-Anderson seismograph, with their product at high and low corner frequency f_0.

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The recorded trace amplitude A may be estimated by noting that a spectrum having Gaussian shape, amplitude $\hat{A}$ and width c yields, on Fourier transformation, a wave packet of amplitude $\hat{A}c/\sqrt{\pi}$. Thus the trace amplitude can be estimated by approximating the product spectrum by a Gaussian shape near its spectral maximum. This approximation is valid only over a small range of fault size R, but the asymptotic behavior at small and large R may be found by considering the schematic spectra shown in Figure 2. The wave displacement spectrum has a corner frequency f_o, say, while the seismograph response spectrum changes from flat at high frequencies to $\sim f^2$ at low frequencies, with a corner frequency f_s. For small faults $f_o \gg f_s$, and the seismograph registers its impulse response multiplied by a factor proportional to the seismic moment; while for large faults $f_o \ll f_s$, and the product spectrum has an amplitude proportional to $(f_o/f_s)^2$ and a nearly constant width $f_s - f_o$. Thus, noting that $M_o \propto \sigma R^3$, we obtain the asymptotic behavior.

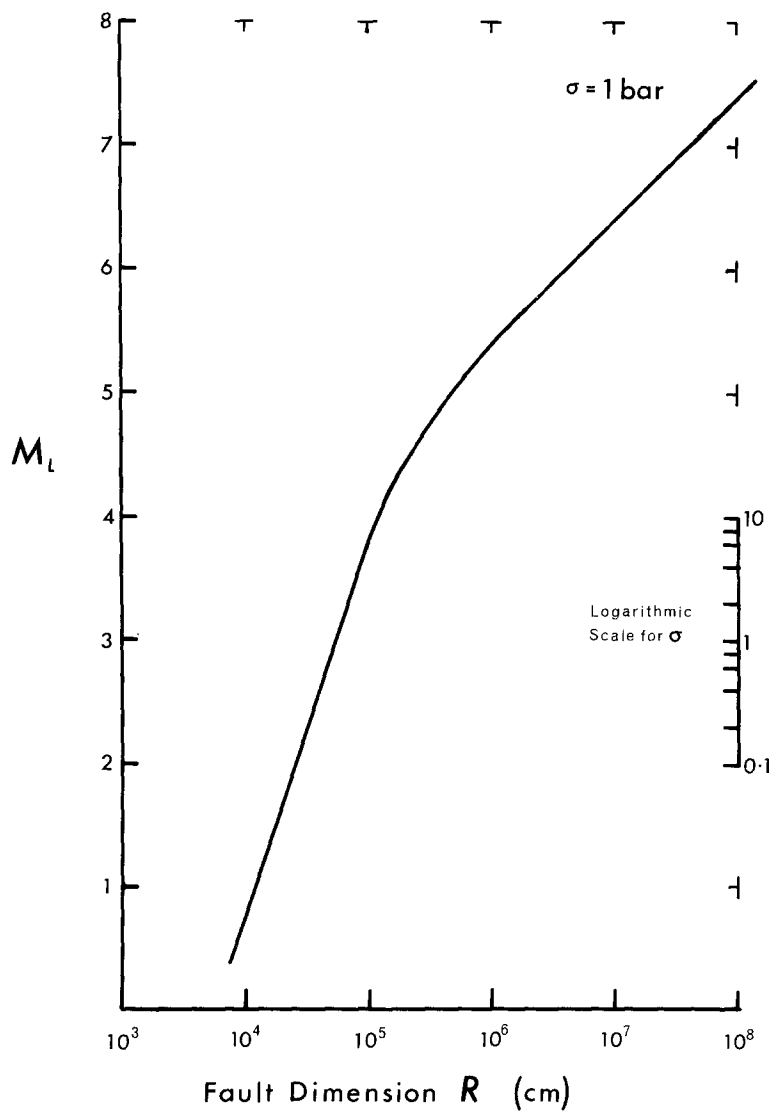

FIG. 3. Theoretical relationship between magnitude M_L and effective fault radius R. The curve for any other stress-drop may be obtained by displacing that for $\sigma = 1$ bar vertically through a distance determined by the logarithmic scale.

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$$10^{M_L} \propto \sigma R^3, \quad R \text{ small};$$

$$10^{M_L} \propto \sigma R, \quad R \text{ large}.$$

The curve for M_L versus R shown in Figure 3 has been obtained by graphically estimating the spectral product and approximating to it a Gaussian shape for the range $R = 1$ to 20 km, and extrapolating to smaller and larger fault radii by using the asymptotic relations. The curve shown is that for $\sigma = 1$ bar. Because M_L is directly proportional to σ, the curve for any other stress-drop is obtained simply by moving the given curve vertically. The shape of the curve is particularly interesting in view of the observation of

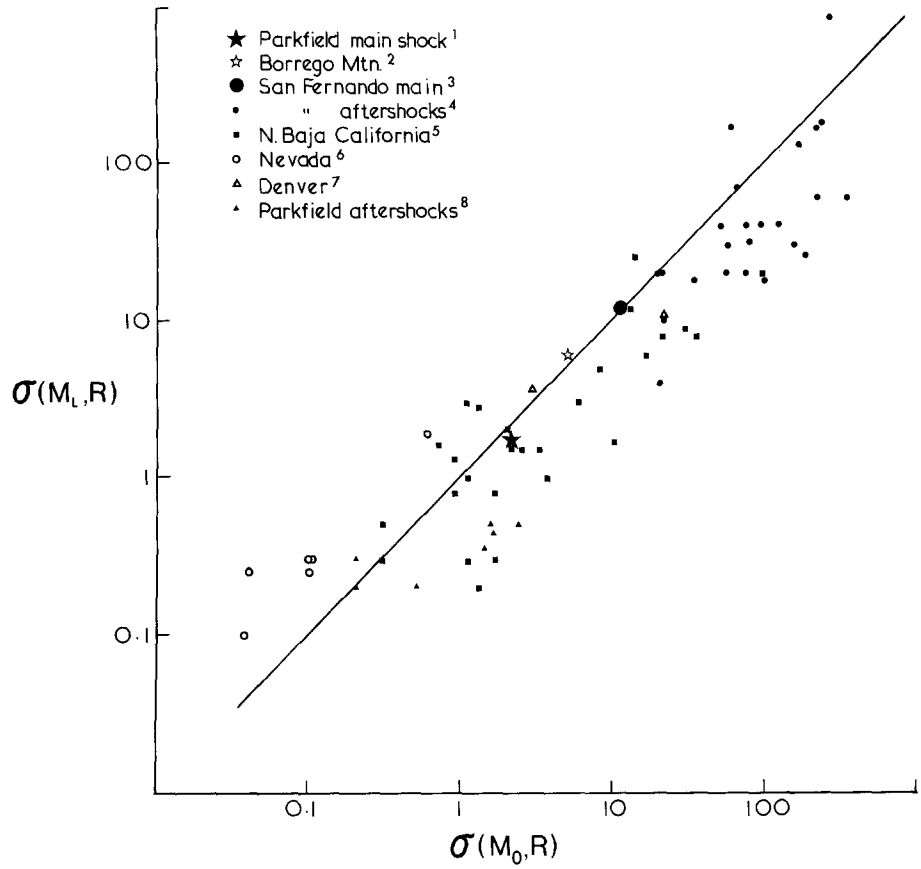

FIG. 4. Comparison of stress-drops calculated from the theoretical relationship between magnitude, fault size and stress-drop, $\sigma(M_L, R)$, with that obtained from spectral estimates of seismic moment and fault size, $\sigma(M_0, R)$. *References*: (1) Aki (1967); (2) Hanks and Wyss (1972); (3) Wyss and Hanks (1972); (4) Tucker and Brune (1973) and Allen *et al.* (1971); (5) Thatcher (1972); (6) Douglas and Ryall (1972); (7) Wyss and Molnar (1972); (8) Wyss and Brune (1968).

Lieberman and Pomeroy (1970) that "There is no doubt that [at smaller magnitudes] the earthquake magnitude begins to fall off more rapidly with decreasing source size than is suggested by the extrapolation of the . . . curves [defined for larger magnitudes]". These authors go on to suggest that "the scatter observed [in M_L verus R diagrams] may be due to differences in effective stress-drop for events in different regions". This would indeed seem to be the case according to the present theory.

Figure 3 may be used to obtain stress-drop from estimates of M_L and R by measuring the height of the corresponding point above the curve for $\sigma = 1$ bar, using the logarithmic scale provided. Figure 4 compares values of stress-drop estimated in this way with

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published values that have been estimated directly from spectral information. The agreement is fairly good over a wide range of stress-drop. This technique is likely to be particularly useful in investigations of large earthquakes followed by aftershock sequences. The extent of aftershock occurrence yields an estimate of source area and hence R [$= (\text{Area}/\pi)^{1/2}$]. Gibowicz (1973) has used this technique to estimate stress-drops for 18 earthquakes in California and New Zealand, and makes useful comparisons between stress-drop and the statistical parameters of their aftershock sequence.

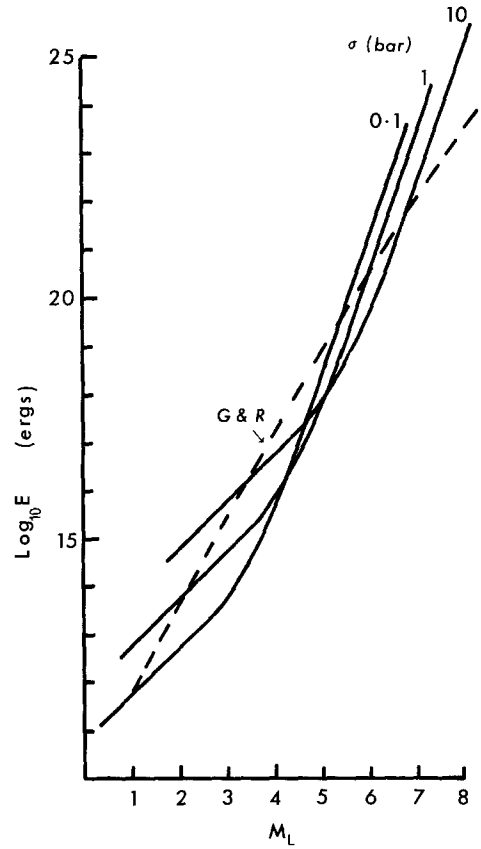

FIG. 5. Theoretical relationship between local earthquake magnitude and seismic energy as a function of stress-drop. The empirical curve of Gutenberg and Richter (1956) is shown dashed.

The spectral theory also implies a relationship between seismic energy and magnitude. Figure 5 shows a family of log E versus M_L curves for $\sigma = 0.1$, 1.0. and 10 bars derived from the theoretical relationship of M_L to σ and R, and from equation (12) for the seismic energy. Also shown is the empirical relationship of Gutenberg and Richter (1956)

$$\log_{10} E = 9.9 + 1.92\, M_L - 0.024\, M_L^2.$$

The theoretical results appear to be consistent with the empirical relationship.

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